Ragini Kalvade

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EDUCATION

University of Illinois, Chicago — Aug 2024 - May 2026
Masters of Science in Computer Science — ***GPA: 4.0/4.0***

National Institute of Technology, India (NIT-B) — May 2018
Bachelor of Technology in Computer Science & Engineering — ***GPA: 3.7/4.0***

EXPERIENCE

Graduate Research Assistant — Jan 2025 - Present
UIC, College of Applied Health Sciences — *Chicago, IL*

- Refined *motion tracking therapy games* using **JavaScript** and **MediaPipe**, deployed on **AWS**; used with **20+ research participants** with multiple sclerosis (MS).
- Created a visual analytics tool to track motor performance, enabling therapists to quantify accuracy, stability, and movement range across **60+ gameplay sessions**.
- Implemented browser-to-server motion data streaming via **WebSockets**, enabling remote physical therapy.

Junior Engineer — Jul 2022 – Jul 2024
Deutsche Bank — *Pune, India*

- Optimized trade pre-booking KYC checks, reducing processing time from **4 days to under 2 minutes.**
- Integrated with **SMTP mail servers** to send transaction booking email and documents created using **Thymeleaf** templates with retry mechanisms to ensure reliable delivery.
- Migrated TradePay infrastructure to **Google Cloud Platform (GCP)**, improving scalability.

Summer Intern — May 2021 – Jul 2021
Deutsche Bank — *Bangalore, IN*

- Streamlined CI/CD pipelines using **Docker, Jenkins, and GCP**; reduced manual effort from **10 minutes to under 2 minutes**.
- Integrated **Elasticsearch**, **cutting log query time by 50%** and improving observability.
- Improved build automation with fixes in dependency and plugin management; **boosted build speed by 80%**.

PROJECTS

Cloud-Based RAG Pipeline for Research Papers | *Scala, Hadoop, MapReduce, Ollama, AWS* — Sept 2025

- Designed and deployed a **cloud-based distributed pipeline** using Hadoop MapReduce to process 600+ research papers, converting PDFs into text chunks and generating embeddings in parallel.
- Built a **Lucene index** of embeddings and integrated with an **LLM (Ollama)** to enable Retrieval-Augmented Generation (RAG) for semantic search across the corpus.

CogniTrack & Fall Prediction Model | *Python, Flask, React, PostgreSQL, MATLAB* — Jun 2025

- Built an interactive analytics dashboard using **Plotly Dash, Flask, and Pandas** to visualize motion data from therapy games — enabled therapists and researchers to conduct longitudinal motor performance analysi.
- Refined data preprocessing pipelines and fixed bugs in a **MATLAB-based fall prediction model** trained on 10+ years of gait and motion tracking data — increased model accuracy and reliability in predicting fall risk.

LitLens: Book Recommender | *GPT-3.5, LangChain, OpenAI API* — Mar 2025

- Trained a conversational AI-powered book recommender using **GPT-3.5**, **LangChain**, and the **OpenAI API**, optimizing prompts to deliver personalized recommendations.
- Leveraged zero-shot classification for dynamic content filtering and improved accuracy.

TECHNICAL SKILLS

Languages & Programming: Python, Java, C++, Go, TypeScript, Bash
Core CS: Data Structures, Algorithms, Object-Oriented Programming, Systems Design
Databases: PostgreSQL, MySQL, MongoDB, Redis, Snowflake
AI & Machine Learning: Supervised/Unsupervised Learning, CNNs, RNNs , LLMs, Natural Language Processing (NLP), Model Evaluation
DevOps & Infrastructure: Docker, Kubernetes, Git, CI/CD Pipelines, GCP, AWS, Azure, ELK Stack
Tools & Frameworks: TensorFlow, PyTorch, scikit-learn, FastAPI, Flask, REST APIs